Prompt 1: Initial Code Generation

Prompt:

Write a beginner-friendly C++ console program that generates random student excuses.
The program should ask the user to enter their name.
Store at least 10 excuse templates using a {name} placeholder.
Randomly select one excuse each time the program runs.
Replace {name} with the actual user input.
Handle empty input properly and display an error message.
Add comments in the code so beginners can understand it.

Code

```
#include <iostream>

#include <vector>

#include <cstdlib>

#include <ctime>

using namespace std;


// BUG: replaces only first occurrence

string personalize(string text, string name) {

   size_t pos = text.find("{name}");

   if (pos != string::npos) {

      text.replace(pos, 6, name);

   }

   return text;

}


int main() {

   string name;

   cout << "Enter your name: ";

   getline(cin, name);
```

```
    if (name.empty()) {
        cout << "Error: Name cannot be empty!" << endl;
        return 0;
    }

    vector<string> excuses = {
        "{name} couldn’t finish because {name}'s laptop crashed.",
        "{name} tried submitting but {name}'s file disappeared.",
        "{name}'s system froze while working.",
        "{name} forgot to save the file.",
        "{name}'s internet stopped working.",
        "{name} accidentally deleted the file.",
        "{name}'s computer restarted suddenly.",
        "{name} faced a power cut.",
        "{name} got logged out suddenly.",
        "{name}'s keyboard stopped working."
    };

    srand(time(0));
    int index = rand() % excuses.size();

    cout << personalize(excuses[index], name) << endl;
    return 0;
}
```

Ouput

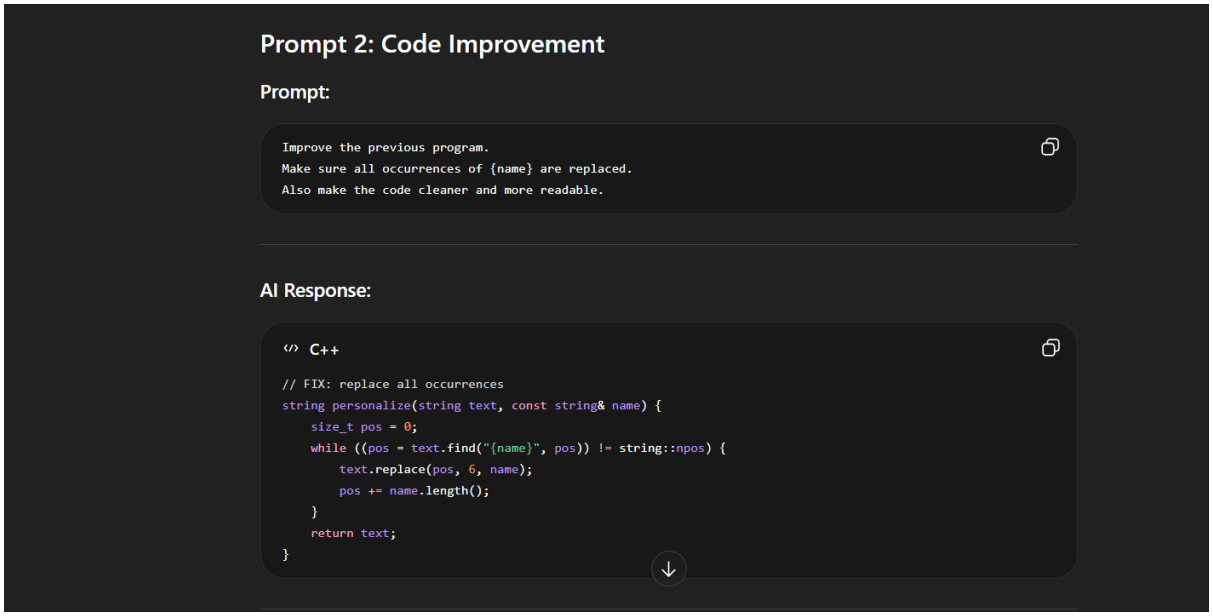
```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                   Code

Warning: PowerShell detected that you might be using a screen reader and has disabled PSReadLine for compatibility purposes. If you want to re-enable it, run 'Import-Module PSRea
dLine'.

PS C:\Users\Nikita\OneDrive\Desktop\focp2-lab manual> cd "c:\Users\Nikita\OneDrive\Desktop\focp2-lab manual\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunnerFile } ; if
($?) { .\tempCodeRunnerFile }
Enter your name: king kong
king kong faced a power cut.
PS C:\Users\Nikita\OneDrive\Desktop\focp2-lab manual>
```

Prompt2:

Improve the previous program.

Make sure all occurrences of {name} are replaced.

Also make the code cleaner and more readable.



Code

```
#include <iostream>

#include <vector>

#include <cstdlib>

#include <ctime>

using namespace std;


// Improved function: replaces ALL occurrences of {name}

string personalize(string text, const string& name) {

   size_t pos = 0;


   // Loop to replace every occurrence

   while ((pos = text.find("{name}", pos)) != string::npos) {

      text.replace(pos, 6, name);

      pos += name.length(); // Move forward after replacement

   }
```

```
    return text;
}

int main() {
    string name;

    // Take input from user
    cout << "Enter your name: ";
    getline(cin, name);

    // Handle empty input
    if (name.empty()) {
        cout << "Error: Name cannot be empty!" << endl;
        return 0;
    }

    // Excuse templates (with multiple {name})
    vector<string> excuses = {
        "{name} couldn’t finish because {name}'s laptop crashed.",
        "{name} tried submitting but {name}'s file disappeared.",
        "{name}'s system froze while {name} was working.",
        "{name} forgot to save the file.",
        "{name}'s internet stopped working when {name} was uploading.",
        "{name} accidentally deleted the file.",
        "{name}'s computer restarted suddenly.",
        "{name} faced a sudden power cut.",
        "{name} got logged out while {name} was submitting.",
        "{name}'s keyboard stopped working during typing."
    };

    // Initialize random seed
```

```
    srand(time(0));

    // Select random excuse
    int index = rand() % excuses.size();

    // Personalize excuse
    string finalExcuse = personalize(excuses[index], name);

    // Display output
    cout << finalExcuse << endl;

    return 0;
}
```

Ouput

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Code

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($?) { .\tempCodeRunnerFile }
Enter your name: priya
priya's internet stopped working when priya was uploading.
PS C:\Users\Nikita\OneDrive\Desktop\focp2-lab manual>
```
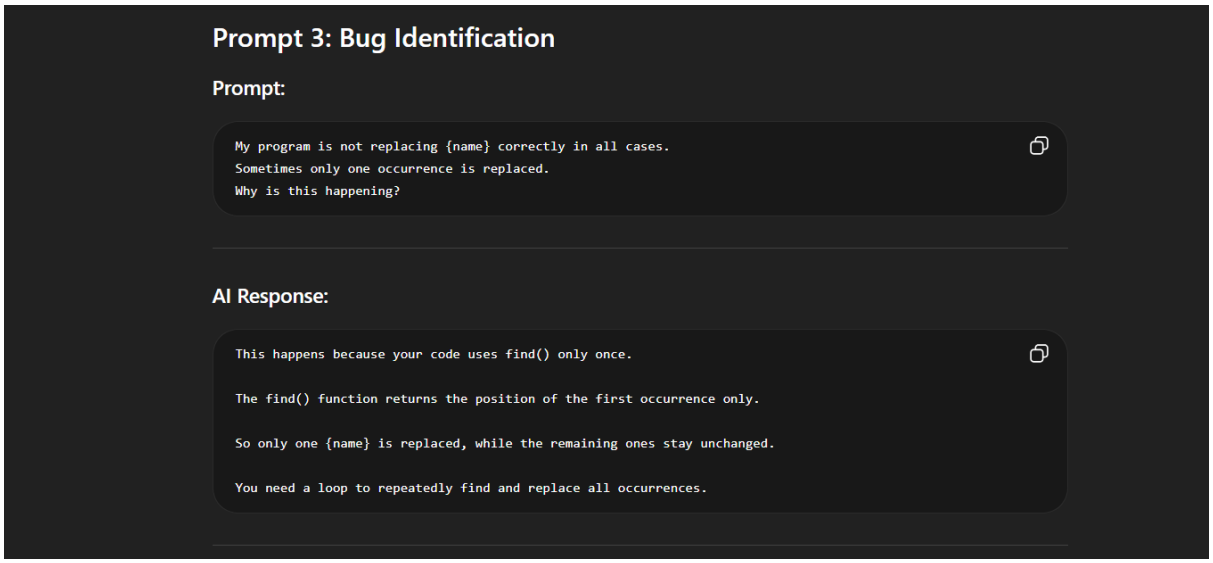

Prompt 3: Bug Identification

Prompt:

My program is not replacing {name} correctly in all cases.
Sometimes only one occurrence is replaced.
Why is this happening?



Prompt 4: Bug Fix Request

Prompt:

Fix the issue so that all occurrences of {name} are replaced correctly.
Explain the fix simply.

Prompt 4: Bug Fix Request

Prompt:

```
Fix the issue so that all occurrences of {name} are replaced correctly.
Explain the fix simply.
```

AI Response:

```cpp
string personalize(string text, const string& name) {
    size_t pos = 0;
    while ((pos = text.find("{name}", pos)) != string::npos) {
        text.replace(pos, 6, name);
        pos += name.length();
    }
    return text;
}
```

Code

```cpp
#include <iostream>
#include <vector>
#include <cstdlib>
#include <ctime>
using namespace std;

// FIXED FUNCTION (from Prompt 4)
string personalize(string text, const string& name) {
   size_t pos = 0;

   // Replace ALL occurrences of {name}
   while ((pos = text.find("{name}", pos)) != string::npos) {
      text.replace(pos, 6, name);
      pos += name.length();
   }

   return text;
}

int main() {
```

```
string name;

// Take user input
cout << "Enter your name: ";
getline(cin, name);

// Handle empty input
if (name.empty()) {
    cout << "Error: Name cannot be empty!" << endl;
    return 0;
}

// Excuse templates
vector<string> excuses = {
    "{name} couldn't finish because {name}'s laptop crashed.",
    "{name} tried submitting but {name}'s file disappeared.",
    "{name}'s system froze while {name} was working.",
    "{name} forgot to save the file.",
    "{name}'s internet stopped working when {name} was uploading.",
    "{name} accidentally deleted the file.",
    "{name}'s computer restarted suddenly.",
    "{name} faced a sudden power cut.",
    "{name} got logged out while {name} was submitting.",
    "{name}'s keyboard stopped working during typing."
};

// Initialize random generator
srand(time(0));

// Select random excuse
int index = rand() % excuses.size();
```

```
    // Apply fix (replace all {name})
    string finalExcuse = personalize(excuses[index], name);

    // Output result
    cout << finalExcuse << endl;

    return 0;
}
```

Output

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Code

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($?) { .\tempCodeRunnerFile }
Enter your name: priya
priya tried submitting but priya's file disappeared.
PS C:\Users\Nikita\OneDrive\Desktop\focp2-lab manual>
```

FINAL CODE

```
#include <iostream>
#include <vector>
#include <cstdlib>
#include <ctime>
using namespace std;

string personalize(string text, const string& name) {
    size_t pos = 0;
    while ((pos = text.find("{name}", pos)) != string::npos) {
        text.replace(pos, 6, name);
        pos += name.length();
    }
    return text;
}

int main() {
```

```
    string name;
    cout << "Enter your name: ";
    getline(cin, name);

    if (name.empty()) {
        cout << "Error: Name cannot be empty!" << endl;
        return 0;
    }

    vector<string> excuses = {
        "{name} couldn’t finish because {name}'s laptop crashed.",
        "{name} tried submitting but {name}'s file disappeared.",
        "{name}'s system froze while working.",
        "{name} forgot to save the file.",
        "{name}'s internet stopped working.",
        "{name} accidentally deleted the file.",
        "{name}'s computer restarted suddenly.",
        "{name} faced a power cut.",
        "{name} got logged out suddenly.",
        "{name}'s keyboard stopped working."
    };

    srand(time(0));
    int index = rand() % excuses.size();

    cout << personalize(excuses[index], name) << endl;
    return 0;
}
```

OUTPUT

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                                                    Code

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($?) { .\tempCodeRunnerFile }
Enter your name: HARSHIT
HARSHIT's internet stopped working.
PS C:\Users\Nikita\OneDrive\Desktop\focp2-lab manual>
```